

# TribalScholar AI — UI/UX Skill Guide

## Purpose

This file tells a designer, frontend engineer, or AI coding agent how to design the TribalScholar AI interface. The product is an AI-assisted, configurable scholarship and fellowship management platform for Scheduled Tribe education schemes. It must feel trustworthy, simple, inclusive, mobile-first, and suitable for low-connectivity environments.

**Core UX promise:** Every applicant should understand what to do next; every officer should understand why the system flagged something; every decision must remain human-authorised and traceable.

## Product principles

1. **Clarity before complexity** — use simple language, short steps, clear labels, and visible progress.
2. **Human decisions, AI assistance** — never present AI results as final rejection, selection, or fraud confirmation.
3. **Explain every important outcome** — show the rule, evidence, reason, deadline, and next action.
4. **Mobile and low bandwidth first** — build for small screens, unreliable networks, inexpensive devices, and slow uploads.
5. **Dignity and fairness** — avoid applicant “trust scores,” stigmatising red flags, or automatic adverse outcomes from spelling/transliteration variation.
6. **Accessible by default** — keyboard support, screen-reader semantics, contrast, large targets, readable typography, and language switching.
7. **Progressive disclosure** — show the next relevant action first; place advanced technical evidence behind expandable details.
8. **Privacy by design** — mask sensitive values, avoid unnecessary collection, and avoid public document links.

## Design system

### Brand direction

- Product name: **TribalScholar AI**
- Tagline: *From application to award—transparent, intelligent and inclusive.*
- Tone: respectful, government-service trustworthy, warm, practical, not overly corporate or futuristic.

- Avoid tribal stereotypes, generic “AI robot” illustrations, surveillance imagery, or imagery that treats applicants as risk subjects.

## Color tokens

Use a calm teal/indigo base with semantic colors. Never use color alone to communicate status.

```
:root {
  --color-primary-900: #073B4C;
  --color-primary-700: #0F6B78;
  --color-primary-600: #0E8A9A;
  --color-primary-100: #E6F6F8;

  --color-accent-700: #5B3F91;
  --color-accent-100: #F0ECF8;

  --color-success-700: #166534;
  --color-success-100: #DCFCE7;
  --color-warning-700: #92400E;
  --color-warning-100: #FEF3C7;
  --color-danger-700: #B42318;
  --color-danger-100: #FEE4E2;
  --color-info-700: #075985;
  --color-info-100: #E0F2FE;

  --color-text: #172B35;
  --color-text-muted: #52616B;
  --color-border: #D7E0E5;
  --color-surface: #FFFFFF;
  --color-canvas: #F6F9FA;
  --color-focus: #155EEF;
}
```

## Typography

- Preferred Latin UI font: Inter, Noto Sans, or system-ui.
- Hindi/Devanagari support: Noto Sans Devanagari or a tested system fallback.
- Base body: 16 px desktop and mobile; never use less than 14 px for essential information.
- Heading scale: 32/28/24/20/18 px, responsive.
- Line height: 1.45–1.65 for body text.
- Use sentence case, not ALL CAPS.
- Use Hindi translations written in clear, everyday terms; avoid bureaucratic wording where possible.

## Layout tokens

```
--space-1: 4px;  
--space-2: 8px;  
--space-3: 12px;  
--space-4: 16px;  
--space-5: 24px;  
--space-6: 32px;  
--space-7: 48px;  
--radius-sm: 8px;  
--radius-md: 12px;  
--radius-lg: 16px;  
--shadow-card: 0 2px 8px rgba(15, 35, 45, .08);
```

- Desktop content max-width: 1200–1280 px.
- Reading/form max-width: 720–800 px.
- Minimum interactive target: 44 × 44 px.
- Use 8 px spacing rhythm.
- Mobile content side padding: 16 px.

## Navigation patterns

### Applicant navigation

Desktop: logo, language switcher, Help, Notifications, Profile menu. Use a clear dashboard side navigation after login.

Mobile: bottom navigation with **Home**, **Applications**, **Help**, and **Profile**. Keep a persistent primary action where appropriate, such as “Continue application.”

Applicant navigation sections:

- Dashboard
- My applications
- Documents
- Notifications
- Help and support
- Profile and settings

### Officer/admin navigation

Use a persistent left sidebar on desktop and a drawer on mobile/tablet.

- Overview
- Review queue
- Applications

- Verification tasks
- Selection/committee
- Schemes and configurator
- Reports and analytics
- Audit logs
- Users and roles
- Settings

Always show current role, jurisdiction/scope, and active scheme context. Example: District Officer • Ranchi • Jharkhand.

Status language and chips

Use text plus icon plus color. Do not call applicants “high risk.” Use review-priority terminology internally and carefully.

| System status          | Applicant-facing label              | Officer-facing label    | Color   |
|------------------------|-------------------------------------|-------------------------|---------|
| DRAFT                  | Draft                               | Draft                   | Neutral |
| SUBMITTED              | Submitted                           | Submitted               | Info    |
| AUTOMATED_CHECK        | Checks in progress                  | Automated check running | Info    |
| DEFICIENCY_RAISED      | Action required                     | Deficiency raised       | Warning |
| RESUBMITTED            | Resubmitted                         | Resubmitted             | Info    |
| INSTITUTE_VERIFICATION | Institute verification pending      | Institute verification  | Info    |
| OFFICER_SCRUTINY       | Under review                        | Officer scrutiny        | Info    |
| COMMITTEE_REVIEW       | Committee review                    | Committee review        | Accent  |
| SELECTED               | Selected, subject to final sanction | Selected                | Success |
| WAITLISTED             | Waitlisted                          | Waitlisted              | Warning |
| REJECTED               | Not selected                        | Rejected                | Danger  |
| SANCTIONED             | Sanctioned                          | Sanctioned              | Success |
| PAYMENT_RELEASED       | Payment released                    | Payment released        | Success |
| MANUAL_REVIEW          | Additional verification required    | Manual review required  | Warning |

Avoid: “fraud,” “suspicious applicant,” “trust score,” “AI rejected,” “invalid person.”

Applicant UX flows

## 1. Registration and OTP

**Goal:** Create an account with minimum friction.

Screen content:

- **Heading:** Create your account
- **Short explainer:** Use your mobile number or email to start an application.
- Mobile or email field.
- Consent link and privacy notice link.
- **Primary CTA:** Send OTP.
- OTP entry has 6 separate accessible inputs or one numeric field, auto-paste support, resend timer, and change-number option.

UX rules:

- Do not block the user with long profile questions before OTP confirmation.
- **Make OTP failures clear:** The code is incorrect or expired. Try again or request a new code.
- For demo/prototype, label OTP as simulated if it is not actually delivered.

## 2. Applicant profile

**Goal:** Collect reusable data once and show why it is needed.

Use sections with save-and-continue behavior:

1. Personal details.
2. Address and location.
3. Category and tribe details.
4. Education details.
5. Income and support details.

Each sensitive field needs helper copy. Example: Your income information is used only to check eligibility for applicable schemes.

Show profile completion card:

Profile completion: 80%  
Next step: Add annual family income to see more preliminary scheme matches.

### 3. Scheme recommendation

**Goal:** Help applicants discover options without making an official eligibility claim.

Card anatomy:

- Scheme name and type.
- “Potential match” label, not “Eligible.”
- 2–4 visible reasons: ST category provided, PhD level selected, Income information available.
- “View requirements” secondary button.
- “Start application” primary button.
- Persistent disclaimer:

This is a preliminary recommendation.  
Final eligibility is decided only after official verification.

Do not hide schemes solely due to incomplete profile. Show Complete profile to check this scheme instead.

### 4. Dynamic application form

**Goal:** Make complex forms feel predictable and finishable.

Use a stepper with 4–6 meaningful stages:

1. Personal details → 2. Education → 3. Scheme details → 4. Documents → 5. Review and

Requirements:

- Save draft automatically and show Saved just now.
- Keep one main column on mobile; use two columns only on large screens for short fields.
- Group fields with headings and explanation text.
- Put required marker in text: Required, not only an asterisk.
- Validate inline after focus leaves the field; summarise errors at the top on submit.
- Preserve typed data after a network interruption.
- Display conditionally required fields only when conditions are met.
- Provide an accessible “Back” action and never discard data without confirmation.

Example field help:

Annual family income  
Enter the total income of your family for the relevant period.  
[ ₹ \_\_\_\_\_ ]

## 5. Document upload

**Goal:** Remove uncertainty around required documents and upload quality.

Display a document checklist, not a generic upload area.

|                     |                      |          |
|---------------------|----------------------|----------|
| ST certificate      | Required             | [Upload] |
| Income certificate  | Required             | [Upload] |
| Academic marksheets | Required             | [Upload] |
| Research proposal   | Required             | [Upload] |
| Admission letter    | Optional/conditional | [Upload] |

Each upload card includes:

- Required/optional label.
- Supported formats and maximum size.
- Sample/helper link only if officially approved.
- Drag/drop on desktop and file picker on mobile.
- Upload progress, retry and replace actions.
- Preview, filename, page count, uploaded time and deletion confirmation.

After OCR:

We found information in your income certificate.  
Annual income: ₹2,40,000  
Does this look correct? [Confirm] [Edit]

If quality issue:

Re-upload recommended  
This file may be blurry or difficult to read. You can continue, but an officer may not

## 6. Review and submit

**Goal:** Prevent accidental submission and make requirements transparent.

Show:

- Completed sections.
- Missing required fields/documents.
- Scheme disclaimer.
- Declaration/consent checkbox.
- Editable links to each section.
- Primary button: Submit application.

Submission confirmation:

Application submitted  
Application number: TS-2026-000124  
We will notify you when verification begins.  
[View application] [Download acknowledgement]

## 7. Applicant dashboard and timeline

Top area:

- Application number and scheme name.
- Current status chip.
- Next required action card when applicable.
- Deadline card when a deficiency is open.

Timeline must be chronological and understandable:

✓ Application submitted – 23 Sep, 11:40 AM  
✓ Automated checks completed – 23 Sep, 11:42 AM  
! Action required: Upload admission letter by 30 Sep  
() Officer verification – starts after you resubmit

For deficiency, show exactly:

- What is missing/wrong.
- Why it matters.
- Deadline.
- Allowed file types.
- Direct Upload document CTA.
- Support route if applicant cannot obtain the document.

## 8. Decision, appeal and grievance

Do not reveal sensitive internal notes. Use respectful, actionable language.

Example rejected-state card:

Decision: Not selected  
Reason: Your application did not meet the configured income criterion after verification.  
You may submit a clarification or appeal by 15 October 2026.  
[Submit clarification] [Raise appeal] [View decision details]

Appeal form fields:

- Application reference, prefilled.
- Category/reason selector.



- Explanation text.
- Supporting-document upload.
- Consent/declaration.
- Submitted grievance number and tracking timeline.

## Officer UX flows

### Review queue

Prioritise tasks without dehumanising applicants.

Table columns:

- Application number.
- Applicant name, masked where not necessary.
- Scheme.
- District/state.
- Current stage.
- Deficiency state.
- Verification priority: Low/Medium/High.
- Assigned officer.
- Last action time.
- SLA/deadline.

Filters:

- Scheme, status, state, district, category, date range, assignee, review priority, document issue, appeal status.

Use a “Why prioritised?” detail drawer. Never show unexplained scores.

### Verification workspace

Three-pane desktop layout:

1. **Application summary:** identity, profile, eligibility results, documents checklist.
2. **Evidence panel:** document viewer, OCR extraction, highlighted fields, quality alerts, cross-document comparison.
3. **Decision panel:** action buttons, required rationale, deficiency template, applicant-visible message preview.

Mobile/tablet: use tabs in this order: Summary → Documents → Evidence → Decision.

AI flag card example:

Possible spelling variation

Confidence: 87%

Evidence: Application name “Laxmi Hembram”; marksheet name “Laxmi Hembrom”

Recommended action: Manual verification

[Accept variation] [Request clarification] [Mark mismatch]

#### Rules:

- Each decision requires a reason and is added to audit log.
- Rejection requires an applicant-visible reason and appeal eligibility check.
- “Accept variation” must record rationale, not simply dismiss AI evidence.
- Provide a clear “view configuration version” link so officers know which rules were used.

#### Merit review and committee screen

- Display configured components and weights.
- Show evidence supporting each component.
- Separate applicant-visible summary from internal committee notes.
- Require committee member identity and timestamp for changes.
- Mark rankings as provisional until approved.
- Show seat count and waitlist boundary.
- Warn when score weights do not total 100 or configuration is unpublished.

#### Admin UX flows

#### Dashboard

Use a calm data-dense layout. Avoid alarming red dashboards unless a critical issue exists.

#### Cards:

- Total applications.
- Pending scrutiny.
- Deficiencies awaiting applicant action.
- Manual review queue.
- Selected candidates.
- Average processing time.
- Payment-release status.

#### Charts:

- Applications by scheme/state/district.
- Pending tasks by officer.

- Deficiency frequency by document type.
- Processing-time trend.
- Selection ratio.
- OCR confidence/quality trend.

Every chart requires a date range and accessible table alternative.

## Scheme configurator

Use a guided builder, not raw JSON as the primary interface. Advanced users may access JSON preview/import/export.

Configuration steps:

1. Basic details
2. Form fields
3. Required documents
4. Eligibility rules
5. Merit and seats
6. Workflow and roles
7. Review and publish

Key interactions:

- Drag-to-reorder fields, with keyboard alternative.
- Rule builder with plain-language preview.
- Validation before publish: deadline validity, required labels, document formats, rule references, score weights total 100, at least one authorised decision role.
- Version comparison before publishing.
- Publish confirmation includes impact warning:

Publishing creates version 3.  
Submitted applications remain linked to their original scheme version.

## Scheme simulator

Inputs:

- Draft income limit.
- Score weights.
- Seats.
- Allowed categories/course levels.

Outputs:

- Count of potentially affected synthetic/authorised records.

- Rank movement summary.
- Warnings about incomplete data.

Use clear label on every result:

Simulation only – this does not change official decisions or published results.

## Fairness dashboard

Show aggregate, privacy-conscious indicators only:

- Deficiency rates by state/district or permitted cohorts.
- OCR error rate by language/script category where lawful and meaningful.
- Manual-review rate for name variations.
- Processing time by location/scheme.
- Selection/rejection disparity signals for human governance review.

Always pair charts with caveat text:

This dashboard identifies patterns for review. It does not establish discrimination or

## Components library

Build reusable components with states: default, hover, focus-visible, disabled, loading, error, empty, success.

- AppShell
- TopBar
- Sidebar
- MobileBottomNav
- LanguageSwitcher
- StatusChip
- ProgressStepper
- FormSection
- FieldHelp
- InlineError
- ErrorSummary
- DocumentChecklist
- DocumentUploadCard
- DocumentViewer
- OCRExtractionCard

- EvidenceComparisonCard
- AIAdvisoryCard
- DeficiencyCard
- Timeline
- AuditTimeline
- MeritScoreBreakdown
- RuleResultCard
- DataTable
- FilterDrawer
- EmptyState
- SkeletonLoader
- ConfirmDialog
- Toast
- AccessibleChart

Microcopy rules

Use direct, respectful wording. Prefer action and explanation.

| Avoid               | Use instead                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Document invalid    | We could not confirm this document. Please upload a clearer income certificate.                                             |
| Fraud suspected     | Additional verification is required because some details need review.                                                       |
| Ineligible          | This application does not meet the configured criterion: [criterion]. You may review the decision or request clarification. |
| Error               | We could not save your changes. Your draft is still available. Try again.                                                   |
| AI decision         | Verification recommendation                                                                                                 |
| High-risk applicant | High verification priority                                                                                                  |

Useful UX messages:

Saved just now  
Your application is still a draft and has not been submitted.  
We will never make a final decision using AI alone.  
This recommendation is preliminary and does not guarantee eligibility.  
Your document is private and can only be accessed by authorised officials.

## Loading, empty and error states

### Loading

- Use skeletons for cards/tables/forms; do not show only a spinning circle for page loads.
- Preserve prior content during filter changes where possible.
- Show upload and OCR stages clearly.

### Empty state examples

No applications yet  
Start by checking schemes that may match your profile.  
[Explore schemes]

No documents require action  
All required documents for this application are currently uploaded.

### Network resilience

- Persist form data locally before API confirmation.
- Show sync state: Saved locally, Syncing, Saved to account, Could not sync.
- Offer retry without losing form data.
- Avoid automatic logout solely because network is unavailable.

### Accessibility requirements

- Meet WCAG 2.2 AA as a target.
- Contrast at least 4.5:1 for normal text and 3:1 for large text/UI components.
- Semantic headings in logical order.
- Native form labels; do not rely only on placeholder text.
- Focus-visible styles on every interactive element.
- Full keyboard access for upload, stepper, menus, tables and dialogs.
- Dialog focus trap and ESC close where safe.
- Announce async updates using polite live regions: upload complete, draft saved, status changed.
- Error summary at top links to invalid fields.
- Do not use color, position, shape or hover alone to convey meaning.
- Charts must have text/table alternatives.
- Support zoom to 200% without horizontal loss of essential content.
- Test with screen reader and Hindi content.

## Responsive requirements

### Mobile: 320–767 px

- Single-column forms.
- Bottom navigation for applicant portal.
- Sticky primary action at bottom only when it does not block field content.
- Turn complex tables into cards or a horizontally scrollable accessible wrapper.
- Document viewer opens full-screen modal.

### Tablet: 768–1023 px

- Two-column dashboard cards.
- Collapsible sidebar.
- Officer workspace uses tabs.

### Desktop: 1024 px and above

- Persistent sidebar for admin/officer users.
- Multi-panel verification workspace.
- Data tables with filter sidebar/drawer.

## UX acceptance criteria

1. A first-time applicant can start and save an application without training.
2. A user always knows current status, next action and relevant deadline.
3. Missing document messages identify the exact document and action.
4. An OCR/name mismatch never appears as an automatic rejection.
5. An officer can see the underlying evidence and record a reason for each decision.
6. An admin can create a new scheme form/document/rule configuration without code changes.
7. Applicant A cannot discover any content belonging to Applicant B through the UI or URL.
8. Every important decision appears in a readable audit timeline.
9. The interface remains usable at 320 px width and at 200% zoom.
10. Every critical flow has loading, success, empty and error states.

## Figma page plan

```
00 Foundations
01 Components
02 Applicant - Auth
03 Applicant - Profile and Recommendations
```

- 04 Applicant - Application Form
- 05 Applicant - Documents and OCR
- 06 Applicant - Dashboard, Deficiency and Appeal
- 07 Officer - Queue and Verification
- 08 Committee - Merit Review
- 09 Admin - Dashboard
- 10 Admin - Scheme Configurator
- 11 Admin - Fairness and Simulator
- 12 Responsive States
- 13 Prototype Flow
- 14 Developer Handoff

## AI builder prompt

Use this UI/UX skill guide to implement the TribalScholar AI frontend.  
Build React + TypeScript components that are mobile-first, accessible and low-bandwidth.  
Use the specified design tokens, status language and component library.  
Implement applicant, officer and admin flows as separate role-based shells.  
Do not present AI flags as final decisions. Every AI advisory must show evidence, context and rationale.  
Render scheme forms and document checklists from backend configuration; do not hard-code.  
Add loading, empty, error, offline-sync and success states for every major flow.  
Use synthetic data only and enforce server-side RBAC through the API.

## Source basis

This UI/UX guide is derived from the uploaded TribalScholar AI concept: configurable scheme management, applicant and officer workflows, OCR/document verification, explainable AI, audit trail, multilingual low-connectivity access, fairness considerations and secure handling of sensitive data.